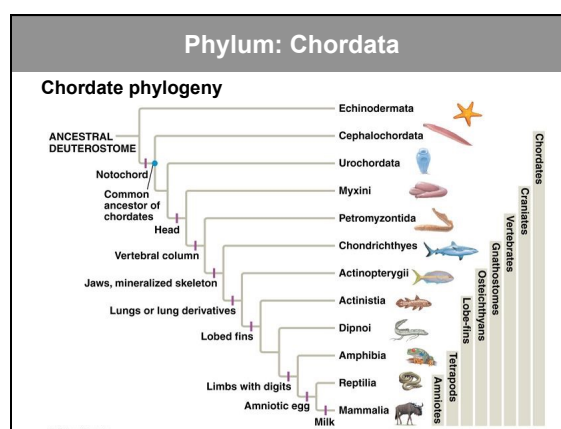
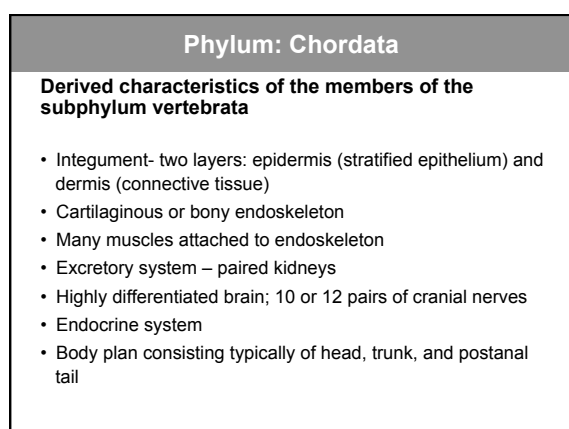
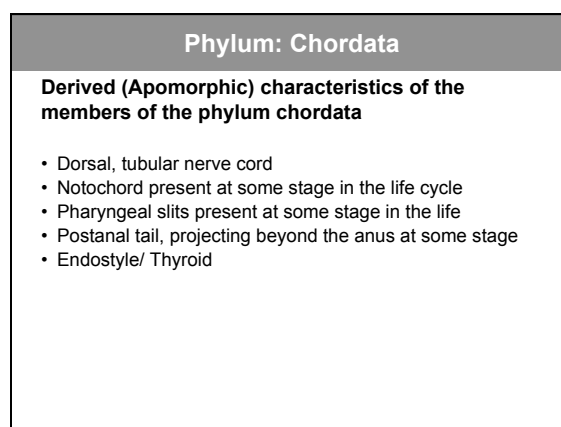
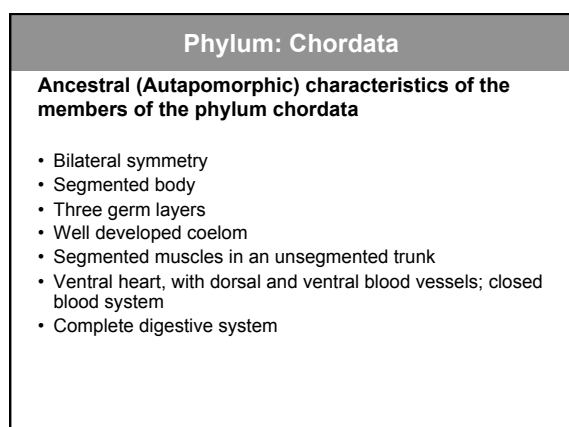
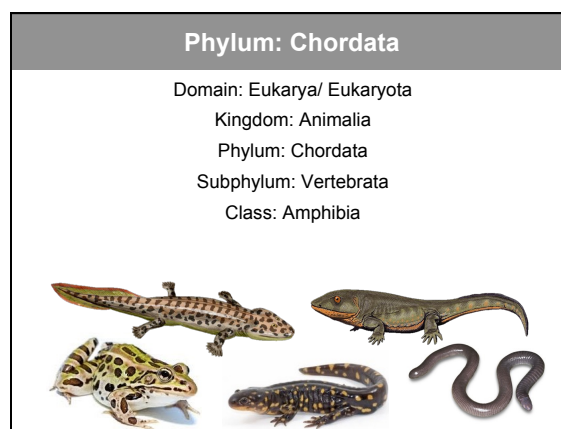




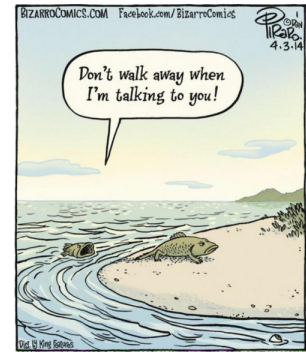
Class: Amphibia

- Commonly called amphibians
- Amphibios - 'amphi' "both kinds", and 'bios' "life"
- Amphibians – anamniotic tetrapod vertebrates
- First vertebrates to colonise terrestrial habitats



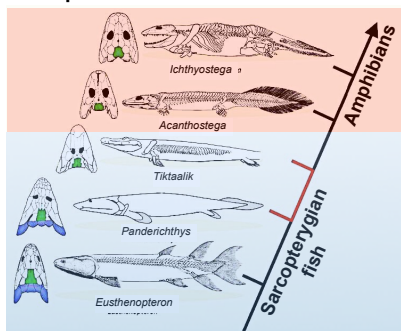
Class: Amphibia

- Earliest amphibians evolved in the Devonian (420-360 Mya) period from sarcopterygian fish
- Amphibians diversified and became dominant during the Carboniferous (360-300 Mya) and Permian (300-250 Mya) periods



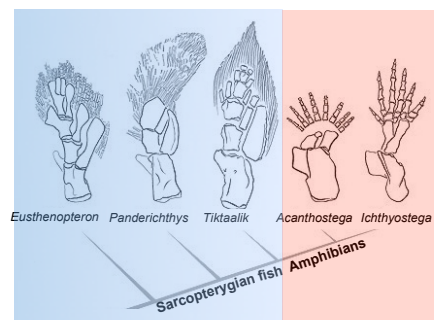
Class: Amphibia

Origins of Amphibians and colonization of Land



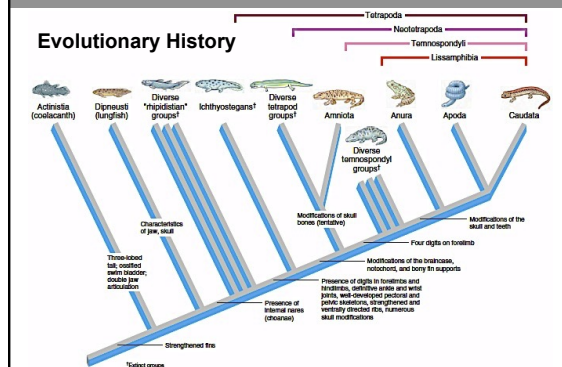
Class: Amphibia

Origins of Amphibians and colonization of Land



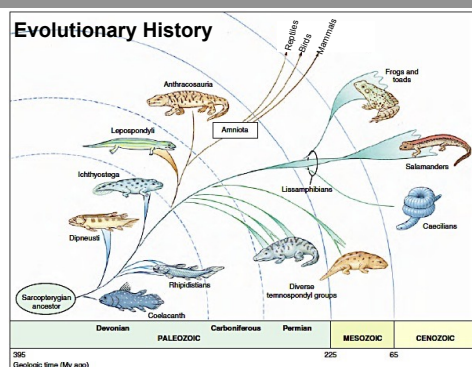
Class: Amphibia

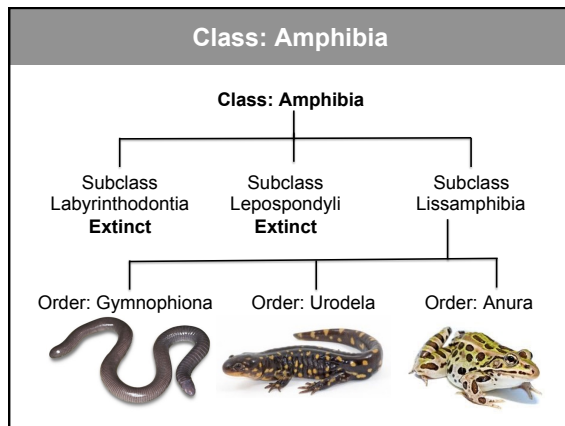
Evolutionary History



Class: Amphibia

Evolutionary History

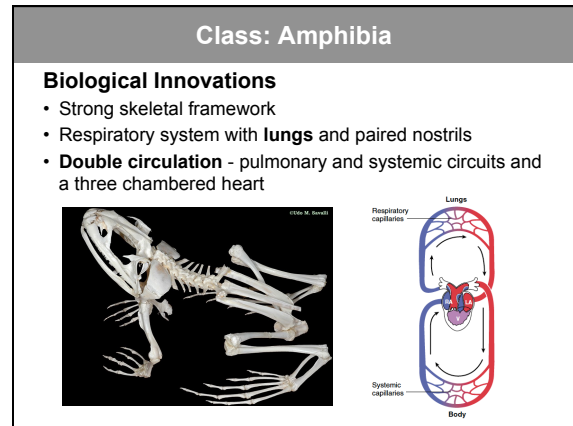




Class: Amphibia

Biological Innovations

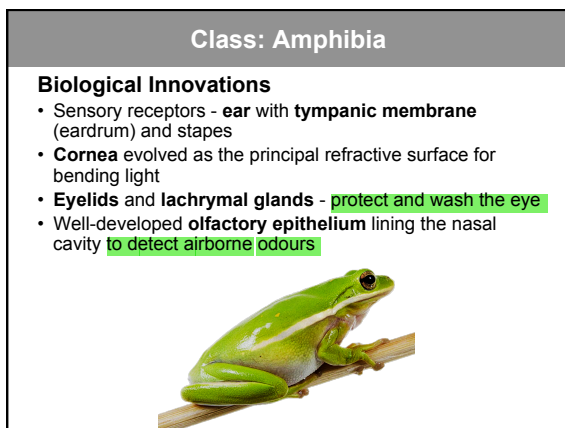
- Strong skeletal framework
- Respiratory system with **lungs** and paired nostrils
- **Double circulation** - pulmonary and systemic circuits and a three chambered heart



Class: Amphibia

Biological Innovations

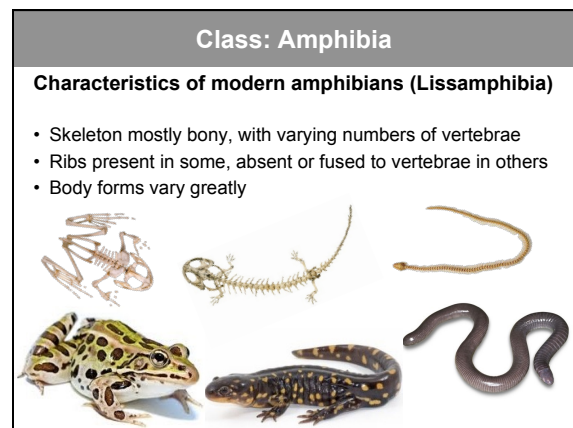
- Sensory receptors - **ear** with **tympanic membrane** (eardrum) and stapes
- **Cornea** evolved as the principal refractive surface for bending light
- **Eyelids** and **lachrymal glands** - **protect and wash the eye**
- Well-developed **olfactory epithelium** lining the nasal cavity **to detect airborne odours**



Class: Amphibia

Characteristics of modern amphibians (Lissamphibia)

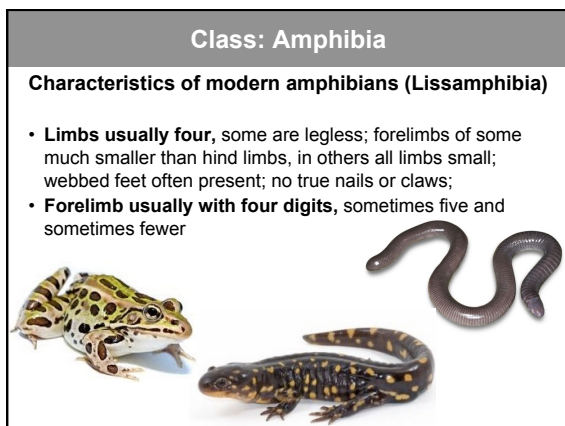
- Skeleton mostly bony, with varying numbers of vertebrae
- Ribs present in some, absent or fused to vertebrae in others
- Body forms vary greatly



Class: Amphibia

Characteristics of modern amphibians (Lissamphibia)

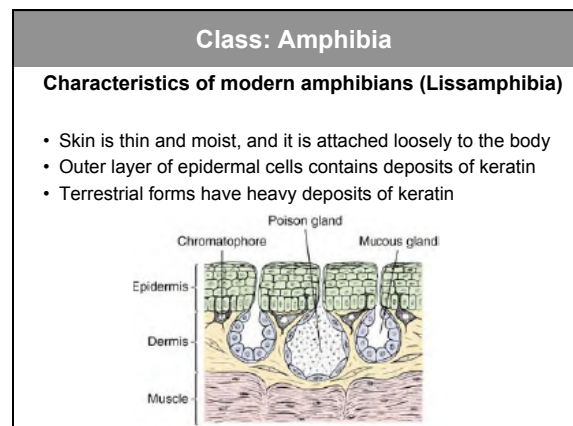
- **Limbs usually four**, some are legless; forelimbs of some much smaller than hind limbs, in others all limbs small; webbed feet often present; no true nails or claws;
- **Forelimb usually with four digits**, sometimes five and sometimes fewer



Class: Amphibia

Characteristics of modern amphibians (Lissamphibia)

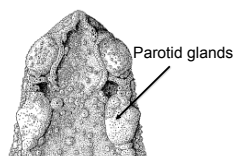
- Skin is thin and moist, and it is attached loosely to the body
- Outer layer of epidermal cells contains deposits of keratin
- Terrestrial forms have heavy deposits of keratin



Class: Amphibia

Characteristics of modern amphibians (Lissamphibia)

- Bears many glands, mostly mucous glands, some may be poison glands
- Pigment cells (chromatophores) common
- No scales, except concealed dermal ones in some

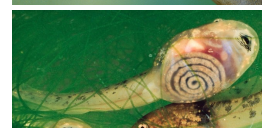
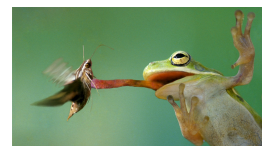


Phylllobates terribilis

Class: Amphibia

Characteristics of modern amphibians (Lissamphibia)

- Adults – carnivores
- Larvae – mostly herbivores, some omnivores, and carnivores
- Adults - Mouth usually large with sticky tongue and small teeth in upper or both jaws
- Adults - digestive tract short
- Larvae – long and coiled



Class: Amphibia

Characteristics of modern amphibians (Lissamphibia)

Respiration

- Two nostrils open into anterior part of mouth cavity
- Respiration: lungs (absent in some), skin and gills in some, either separately or in combination
- Larval stages use external gills and in some gills may persist throughout life in some

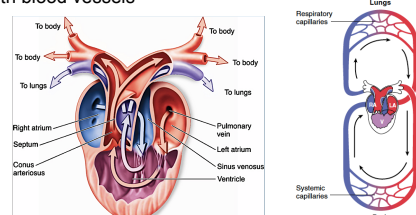


Class: Amphibia

Characteristics of modern amphibians (Lissamphibia)

Circulation

- Circulation with three-chambered heart - two atria and one ventricle
- Double circulation through the heart, skin abundantly supplied with blood vessels

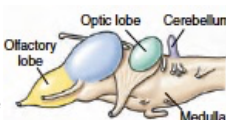


Class: Amphibia

Characteristics of modern amphibians (Lissamphibia)

Nervous system

- Brain – Forebrain, midbrain, hindbrain
- Forebrain (telencephalon) – sense of smell (contains the olfactory center)
- Midbrain (mesencephalon) - vision
- Hindbrain (rhombencephalon) – hearing and balance
- Cerebrum (part of the forebrain) – poorly developed
- Ten pairs of cranial nerves

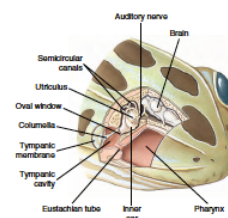


Class: Amphibia

Characteristics of modern amphibians (Lissamphibia)

Sensory organs

- Well developed hearing in anurans
- Single earbone (i.e. columella/stapes)
- Sensitive to high and low frequency sounds (<4000 Hz)

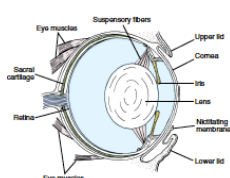


Class: Amphibia

Characteristics of modern amphibians (Lissamphibia)

Sensory organs

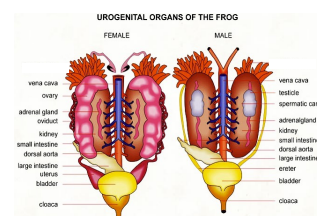
- Well developed vision in anurans
- Retina contains both rods and cones
- Iris contains well developed circular and radial muscles and can rapidly expand or contract pupil) to adjust to changing light
- Upper eyelid is fixed
- Lower eyelid is folded into a transparent membrane – nictitating membrane



Class: Amphibia

Characteristics of modern amphibians (Lissamphibia) Excretion

- Excretory system – paired mesonephric kidney
- Nitrogenous waste – Urea (main)



Class: Amphibia

Characteristics of modern amphibians (Lissamphibia)

Thermoregulation

- Ectothermal
- Behavioural thermoregulation – inactive during highly warm and very cold temperatures
- Absent in regions where the temperature is very cold all year

Class: Amphibia

Characteristics of modern amphibians (Lissamphibia) Reproduction

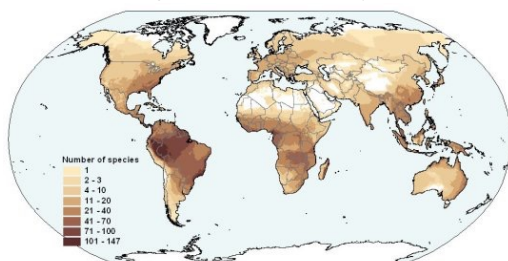
- Sexes – Separate
- Fertilization - mostly internal in salamanders and caecilians
- mostly external in frogs and toads
- Oviparous
- Metamorphosis usually present
- Eggs - Moderately yolky with jellylike membrane



Class: Amphibia

Characteristics of modern amphibians (Lissamphibia) Distribution

Distribution of global amphibian diversity: ~8343 species



Class: Amphibia

Summary

- Amphibians are anamniotic tetrapod vertebrates
- First vertebrates to colonise terrestrial habitats
- Evolved in the Devonian (420-360 Mya) period from sarcopterygian fish
- Only the members of the subclass Lissamphibia are existence today
- Lissamphibia consists of the three orders: Anura, Gymnophiona, Urodela
- Their biological Innovations include; strong skeletal framework, lungs, double circulation, tympanic membrane, eyelids, lachrymal glands and olfactory epithelium
- Skink consists of epidermis and dermis
- Adults are carnivorous and larvae are herbivorous

Class: Amphibia

Summary

- Respiratory surfaces: lungs, skin, gills (in larvae)
- Hear consists only of three chambers
- Brain – Forebrain, midbrain, hindbrain
- Single earbone (stapes)
- Excretory products is Urea
- Ectothermic animals
- Reproduction is sexual, larval stage undergoes metamorphosis
- Distributed all over the world except Antarctica